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Do urinary BMP levels change with disease progression in idiopathic and genetic PD?

Kalpana Merchant, TransThera Consulting Co.
Frank Hsieh, Nextcea, Inc.

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Summary

Urine biospecimens from the LRRK2 genetic cohort of PPMI have been analyzed for three phospholipids: total di-BMP-22:6, its distinct 2, 2'-isoform (2, 2' di-22:6-BMP)and total di-BMP-18:1. The following table lists the biospecimens analyzed.

Study Arm	**BL Visit**	**V04 (1 year)**	**V06 (2 year)**	
LRRK2 PD	150	121	82	
LRRK2 Unaffected	202	159	100	
Total	**352**	**280**	**182**	**866**

Method

Analysis of BMP isoforms:

Quantitative Ultra Performance Liquid Chromatography Tandem Mass Spectrometry (UPLC-MS/MS) analyses: Urinary BMPs were extracted by liquid-liquid extraction and measured by targeted UPLC-MS/MS as described previously by Liu et al., 2014. Absolute levels of total di-22:6-BMP (the sum of its three isoforms), its 2, 2'-isoform (2,2'-di-22:6-BMP), and total di-18:1-BMP were measured using a SCIEX X500B QTOF LC-MS/MS system (SCIEX, Framingham, MA). Injections were made using a Shimadzu Nexera XR UPLC system (Shimadzu Scientific Instruments, Japan). The instruments were controlled by SCIEX OS Software. Quantitation was performed using authentic di-22:6-BMP and di-18:1-BMP reference standards. Di-14:0-BMP was used as an internal standard.

Calibration and data processing: The intensities of the analytes and internal standards were determined by integration of extracted ion peak areas using SCIEX OS-Q Software. Calibration curves were prepared by plotting the peak area ratios for each analyte to internal standard versus concentration. The model for the calibration curve was linear with ($1/x^2$) weighting. Measured concentrations of the urinary lipids (ng/mL) were normalized by the concentration of urine creatinine and reported in ng/mg creatinine.

Creatinine measurement:

Concentrations of urinary creatinine were measured by colorimetric assay (method of Jaffé) with Parameter Creatinine Assay test reagents (R&D Systems, Minneapolis, MN) using a BioTek ELx800 absorbance microplate reader with Gen5 Microplate Reader and Imager Software 2.09 (Fisher Scientific, Hampton, NH).

References

Liu N, Tengstrand EA, Chourb L, Hsieh FY. Di-22:6 bis(monoacylglycerol)phosphate: A clinical biomarker of drug-induced phospholipidosis for drug development and safety assessment. Toxicology and applied pharmacology 2014;279:467-476).

About the Authors

This document was prepared by Kalpana Merchant (TransTHera Consulting Co.) and Frank Hsieh (Nextcea, Inc.). For more information please contact Kalpana Merchant at merchant.transthera@gmail.com or Frank Hsieh at frank.hsieh@nextcea.com